

Enterprise Infrastructure

Part 1 – Company Profile

1. Company Name

ABC Startup Solutions

ABC Startup Solutions is a newly established software development and IT solutions company that specializes in creating modern digital solutions for businesses and organizations.

2. Nature of Business

ABC Startup Solutions is a technology company that focuses on software development, business systems, web applications, database management, cloud solutions, and various IT services.

The company creates customized digital solutions that allow organizations to automate their operations, efficiently manage information, increase productivity, and improve the quality of services provided to their customers.

3. Company Vision

To establish ABC Startup Solutions as a trusted technology provider that helps transform businesses through innovative, secure, and dependable digital solutions.

4. Company Mission

ABC Startup Solutions aims to create practical and innovative software solutions that enable organizations to improve their operations, productivity, and customer experience. The company is committed to providing secure, dependable, and user-centered technologies while continuously enhancing its services to address the changing needs of its clients.

5. Office Location

ABC Startup Solutions

2nd Floor, ABC Startup Solutions

National Highway, Calamba, Laguna, Philippines

The company will occupy one office floor that can accommodate its 20 employees. The workplace will include designated areas for each department, a server and network area, meeting rooms, and shared facilities.

The company's IT infrastructure will be centrally managed to ensure employees have dependable internet connectivity, secure access to internal resources, file storage, communication tools, and business applications.

6. Organizational Structure

ABC Startup Solutions will follow a **functional organizational structure**, where employees are assigned to departments according to their responsibilities, skills, and areas of specialization.

Department Responsibilities

Information Technology Department

- Develops and maintains the company's software systems.
- Manages servers, computers, networks, and other IT infrastructure.
- Provides technical assistance to employees.
- Maintains cybersecurity and system availability.

Human Resources Department

- Manages recruitment and employee onboarding.
- Maintains employee records.
- Handles company policies and employee-related procedures.
- Supports employee development.

Finance Department

- Manages accounting and financial records.
- Handles payroll and company expenses.
- Prepares financial reports.
- Supports budgeting and financial planning.

Sales Department

- Manages customer acquisition and sales activities.
- Maintains relationships with clients.
- Identifies potential business opportunities.
- Coordinates with the IT department regarding customer requirements.

7. Employee Distribution

ABC Startup Solutions will begin operations with **20 employees** distributed across four departments. Based on the proposed organizational setup, there will be **5 employees in Information Technology, 4 in Human Resources, 5 in Finance, and 6 in Sales.**

Department	Employees	Primary Responsibilities
Information Technology	5	Software development, infrastructure, system administration, and technical support
Human Resources	4	Recruitment, employee records, and administration
Finance	5	Accounting, payroll, budgeting, and financial management
Sales	6	Sales, customer relations, and business development
Total	20	Overall company operations

Proposed Positions

Department	Position	Number
Information Technology	IT/System Administrator	1
Information Technology	Software Developers	3
Information Technology	IT Support Technician	1
Human Resources	HR Manager	1
Human Resources	HR Officers	2
Human Resources	HR Assistant	1
Finance	Finance Manager	1
Finance	Accountants	2
Finance	Finance Officers	2
Sales	Sales Manager	1
Sales	Sales Representatives	4
Sales	Business Development Officer	1
Total		20

Company Profile Summary

ABC Startup Solutions is a newly established software development and IT solutions startup with **20 employees**, operating from a single office floor in Calamba, Laguna. The company offers software development, business systems, database solutions, cloud services, and technical support to its clients.

The organization is divided into four main departments: **Information Technology, Human Resources, Finance, and Sales**. The IT department is responsible for developing the company's technology solutions and maintaining its internal IT infrastructure.

Since ABC Startup Solutions currently does not have computers, servers, networking equipment, internet infrastructure, or established security policies, a complete IT infrastructure plan is necessary before the company begins its operations.

This infrastructure plan will establish a dependable foundation for the company's current activities while also providing the flexibility needed to support future growth in employees, services, and technology.

Part 2 – Enterprise Hardware Inventory

ABC Startup Solutions requires a range of hardware components to support the daily activities of its 20 employees. The selected equipment is based on the needs of each department and the company's IT operations.

Hardware	Quantity	Department	Purpose
Desktop Computers	15	IT, HR, Finance, Sales	Primary computers for employees' daily office tasks
Laptops	5	IT, Sales	Portable devices for work, presentations, and meetings
Server	1	IT	Stores company files, applications, and essential services
Router	1	IT	Connects the office network to the internet
UPS	3	IT	Protects important equipment from sudden power interruptions
Wireless Access Point	2	IT	Provides wireless network coverage throughout the office
NAS Storage	1	IT	Stores shared files and backup data
External Backup Drives	2	IT	Provides additional copies of important company data
Monitors	20	All Departments	Provides a display for each employee
Managed Switch	1	IT	Connects computers and other wired network devices

Hardware	Quantity	Department	Purpose
Network Printer	2	HR, Finance	Used for printing documents, reports, and other office materials

The hardware components were selected according to the operational requirements of ABC Startup Solutions and its 20 employees. Sufficient computers are needed to allow employees to perform their daily responsibilities, while additional equipment is required by the IT department to manage the company's systems and network.

The **server and NAS storage** will provide centralized storage for company files, applications, and important data. Meanwhile, the **router, managed switch, and wireless access points** will establish the network and internet connectivity required by employees.

Two **network printers** are included for departments that regularly handle documents and reports. The **UPS units** will help protect critical equipment from unexpected power interruptions, while the **external backup drives** will provide additional copies of important files for data recovery.

Part 3 — Enterprise Software Inventory

ABC Startup Solutions will use a combination of operating systems, productivity applications, development tools, server software, and security programs to support the company's daily operations and IT requirements.

Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	Current	Paid	Primary operating system for employee computers
Ubuntu Server	24.04 LTS	Free / Open Source	Used for the company server and IT-related services
Microsoft 365	Current	Subscription	Used for documents, spreadsheets, presentations, and other office activities
Visual Studio Code	Current	Free	Used by developers for writing and editing source code
Git	Current	Free / Open Source	Used for tracking and managing changes to source code
GitHub Desktop	Current	Free	Provides an easier way to manage Git repositories
VirtualBox	Current	Free / Open Source	Used to create and test virtual machines
Google Chrome	Current	Free	Main web browser for accessing websites and online services

Software	Version	License	Purpose
Microsoft Defender	Built-in	Included with Windows	Provides basic antivirus and security protection
AnyDesk	Current	Free / Paid	Used by IT personnel for remote technical assistance
7-Zip	Current	Free / Open Source	Used for compressing and extracting files

The selected software applications are based on the different responsibilities and activities of ABC Startup Solutions employees. **Windows 11 Pro** and **Microsoft 365** will support regular office operations, while **Visual Studio Code**, **Git**, and **GitHub Desktop** will provide the necessary tools for software developers.

The company's **Ubuntu Server** will be used to operate the server and provide required IT services. The IT department can use **VirtualBox** to create virtual environments for testing operating systems, applications, and different configurations without affecting the main systems.

Google Chrome will serve as the primary browser for accessing websites, web-based applications, and online services. For basic endpoint security, **Microsoft Defender** will provide antivirus and malware protection on Windows computers.

For technical assistance, **AnyDesk** will allow IT personnel to remotely access computers when providing support. Meanwhile, **7-Zip** will be useful for compressing and extracting files when handling software packages, documents, and other data.

Overall, the proposed software inventory provides ABC Startup Solutions with the essential applications needed for office productivity, software development, server management, IT support, testing, file management, and basic security.

Part 4 — Enterprise Network Inventory

ABC Startup Solutions will require several networking components to establish a reliable and secure network for its 20 employees. The selected equipment will provide internet connectivity, connect wired and wireless devices, organize network cabling, and help protect the company's internal network.

Networking Equipment

Networking Equipment	Quantity	Department/Location	Purpose
ISP Modem	1	Server Room	Provides the internet connection from the Internet Service Provider
Router	1	Server Room	Manages communication between the internet and the company's internal network

Networking Equipment	Quantity	Department/Location	Purpose
Firewall	1	Server Room	Helps protect the network from unauthorized access
Managed Switch	1	Server Room	Connects wired computers and other network devices
Wireless Access Point	2	Office Area	Provides wireless network connectivity throughout the office
Patch Panel	1	Server Room	Organizes and connects network cables
CAT6 Ethernet Cable	300 meters	Whole Office	Connects computers and other network devices
RJ45 Connectors	50 pieces	IT	Used to create and maintain Ethernet cable connections

The proposed networking equipment is sufficient for the initial setup of ABC Startup Solutions' 20-employee office. The **ISP modem and router** will provide the company's internet connection and manage communication between the external internet and internal network.

The **firewall** will provide an additional layer of protection by helping prevent unauthorized access to the company's network. The **managed switch** will be responsible for connecting wired computers, servers, printers, and other network devices.

Two **wireless access points** will provide Wi-Fi coverage across the office, allowing employees to connect their laptops and other wireless devices. The **patch panel** will help keep network connections organized, making the cabling system easier to manage and troubleshoot.

Overall, these networking components provide ABC Startup Solutions with the basic infrastructure required for internet access, wired and wireless connectivity, network organization, and security while allowing the network to accommodate future expansion.

Part 6 — System Administration Roles

Helpdesk Technician

Responsibilities

- Assists employees in resolving computer and software-related issues.
- Installs and sets up basic software applications.
- Troubleshoots common hardware and network problems.
- Refers more complex technical issues to the appropriate IT personnel.

Skills

- Technical troubleshooting
- Communication
- Basic networking
- Hardware and software knowledge

Common Tools

- AnyDesk
- Windows tools
- Ticketing systems
- Remote support applications

Certifications

- CompTIA A+
 - Google IT Support Certificate
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Network Administrator

Responsibilities

- Manages and maintains the company's network infrastructure.
- Configures routers, switches, and wireless access points.
- Monitors network performance and connectivity.
- Assists in maintaining network security.

Skills

- Computer networking
- IP addressing
- Network troubleshooting
- Network security

Common Tools

- Cisco Packet Tracer
- Wireshark
- Command Prompt
- Router and switch management tools

Certifications

- CompTIA Network+
 - Cisco CCNA
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Linux System Administrator

Responsibilities

- Manages and maintains Linux-based servers.
- Installs, configures, and maintains server software.
- Manages user accounts and access permissions.
- Monitors server performance and security.

Skills

- Linux commands
- Server administration
- Shell scripting
- Troubleshooting

Common Tools

- Ubuntu Server
- Bash
- SSH
- VirtualBox

Certifications

- Linux+
 - LPIC-1
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Cloud Administrator

Responsibilities

- Manages the company's cloud-based services.
- Creates, configures, and maintains cloud resources.
- Monitors cloud system performance and availability.

- Helps maintain cloud security and access controls.

Skills

- Cloud computing
- Networking
- Security
- System administration

Common Tools

- AWS
- Microsoft Azure
- Cloud management consoles
- Virtual machines

Certifications

- AWS Certified Cloud Practitioner
- Microsoft Azure Fundamentals (AZ-900)

How They Work Together

The four IT professionals collaborate to ensure that ABC Startup Solutions' technology systems operate effectively. The **Helpdesk Technician** handles common technical concerns from employees and forwards more complex problems to the appropriate IT personnel.

The **Network Administrator** is responsible for managing network connectivity and infrastructure, while the **Linux System Administrator** oversees the company's Linux server and related services. Meanwhile, the **Cloud Administrator** manages the organization's cloud resources and services.

By working together and coordinating their responsibilities, these IT professionals can resolve technical issues efficiently, maintain system availability, and help keep the company's technology infrastructure secure and reliable.

Part 7 — Infrastructure Recommendations

Internet Provider

ABC Startup Solutions should subscribe to a **reliable business internet plan with a minimum speed of 300 Mbps**. This connection should adequately support the 20 employees who require internet access for software development, communication, cloud-based services, and other daily business activities.

Server Specifications

The company should acquire a server with, at minimum, the following specifications:

- Intel Xeon or an equivalent server-grade processor
- 32 GB RAM
- 2 TB storage capacity
- RAID storage configuration
- Gigabit Ethernet
- Ubuntu Server

These specifications should be sufficient to support the company's current operations. As the organization expands, the server can be upgraded with additional resources when necessary.

Backup Strategy

Important company files should be backed up regularly to the **NAS storage**. Another copy of the data should also be maintained using the external backup drives.

The IT department should establish a regular backup schedule and periodically verify that stored backups can be successfully restored. This will help ensure that important information can be recovered if data is accidentally deleted, corrupted, or lost.

Security Recommendations

ABC Startup Solutions should implement several basic security measures, including a firewall, strong passwords, user access controls, and regular software updates. Employees should also be instructed to avoid suspicious links, unknown downloads, and potentially harmful files.

Access to sensitive company information should be limited to employees who require the data to perform their assigned responsibilities.

Antivirus

Microsoft Defender should remain enabled on all Windows-based computers. It provides basic protection against viruses, malware, and other common security threats that could affect employee devices.

Password Policy

Employees should create strong and unique passwords for their accounts. Passwords must not be shared with other employees and should be changed whenever there is a possibility that an account has been compromised.

Multi-factor authentication (MFA) should also be enabled for important accounts whenever the service supports it. This provides an additional layer of protection beyond the user's password.

Expansion Plan

As ABC Startup Solutions increases its workforce and operations, the company can expand its infrastructure by adding additional computers, network ports, storage capacity, and wireless access points. The existing server can also be upgraded with additional RAM and storage as required.

The network should be designed with additional capacity from the beginning so that new employees and devices can be accommodated without requiring a complete redesign or replacement of the existing infrastructure.

Overall Recommendation

The proposed infrastructure is appropriate for ABC Startup Solutions as a small and growing startup. It provides the necessary computing equipment, network resources, security measures, and backup capabilities required to support the company's current 20 employees.

At the same time, the proposed setup allows room for future expansion, enabling the company to increase its workforce, equipment, storage, and network capacity as its operations grow.

Part 8 — Personal Reflection

Working on this project helped me realize that system administration involves more than simply repairing computers or managing networks. It also requires careful planning of an organization's IT infrastructure before the systems are implemented. Through this project, I learned the importance of considering the number of employees, their assigned departments, their responsibilities, and the hardware and software required to support their work.

One of the important lessons I gained from this project was how to prepare hardware, software, and network inventories. Before completing this project, I mainly viewed computers and networking as separate components. However, I now understand that these elements must work together to support the overall operations of an organization. I also learned that different IT roles, such as helpdesk technicians, network administrators, Linux system administrators, and cloud administrators, have different responsibilities while still needing to collaborate with one another.

The most challenging part of the project for me was planning the network infrastructure. I needed to determine how different components, including the internet connection, modem, router, firewall, switch, server, wireless access points, and printers, would connect and communicate across the different departments. Designing the network diagram helped me better understand how the infrastructure of an actual office environment can be organized.

This project can help me develop my skills as a future System Administrator because it allowed me to practice identifying and planning an organization's IT requirements before implementation. It also improved my abilities in technical documentation, network planning, and problem-solving. I learned that a capable System Administrator should not only be able to troubleshoot existing problems but should also know how to properly plan, organize, and maintain IT systems.

Overall, this project provided me with a better understanding of how IT infrastructure is planned and organized within a real-world organization. It demonstrated the importance of proper preparation, documentation, security, and resource management in developing an IT environment that is reliable, secure, manageable, and capable of supporting future growth.